Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Dairy Central

Physical address of dairy:

Number and Street	City	County	Zip Code
6626 N Central AVE	Hilmar	Merced	95324

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1932

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X050-X002-XXXX

B. OPERATORS

Borba, Bob

Operator name: Borba, Bob

Telephone no.: (209) 321-5444 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6480 Mitchell RD	Hilmar	CA	95324

This operator is responsible for paying permit fees.

C. OWNERS

Borba, Bob

Legal owner name: Borba, Bob

Telephone no.: (209) 321-5444 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
6480 Mitchell RD	Hilmar	CA	95324

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	1,600	250	0	0	0	0
Maximum number	1,600	250	0	0	0	0
Average number	1,600	250	0	0	0	0
Avg live weight (lbs)	1,400	1,500	0	0		

Predominant milk cow breed: Holstein

Average milk production: 77 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 46,067.69 *tons per reporting period*

Total nitrogen from manure: 599,419.11 *lbs per reporting period*

After ammonia losses (30% loss applied): 419,593.38 *lbs per reporting period*

Total phosphorus from manure: 100,027.06 *lbs per reporting period*

Total potassium from manure: 313,773.16 *lbs per reporting period*

Total salt from manure: 810,847.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 12,833,884 *gallons*

Total nitrogen generated: 200,018.93 *lbs*

Total phosphorus generated: 92,552.52 *lbs*

Total potassium generated: 149,589.43 *lbs*

Total salt generated: 403,351.89 *lbs*

	12,833,884	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	12,833,884	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Domestic Well - DW #1	Ground water
Irrigation Water - Canal	Surface water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	80	80	7	process wastewater	X045-X050-X001-XXXX
10	28	28	2	process wastewater	X045-X080-X020-XXXX
11	17	17	7	process wastewater	X045-X080-X020-XXXX
12	10	10	7	process wastewater	X045-X080-X020-XXXX
13	14	14	7	process wastewater	X045-X050-X006-XXXX X045-X080-X020-XXXX
2	12	12	7	process wastewater	X045-X050-X002-XXXX
3	7	7	7	process wastewater	X045-X050-X002-XXXX
4	40	40	7	process wastewater	X045-X050-X002-XXXX
5	22	22	2	process wastewater	X045-X050-X014-XXXX
6	68	68	2	process wastewater	X045-X050-X017-XXXX
7	84	84	2	process wastewater	X045-X050-X016-XXXX X045-X050-X021-XXXX
8	23	23	2	process wastewater	X045-X050-X017-XXXX
9	21	21	2	process wastewater	X045-X050-X017-XXXX
Fowler	20	20	2	manure	X045-X032-X012-XXXX
Harding	38	38	2	manure	X044-X047-X003-XXXX
Totals for areas that were used for application	484	484	65		
Totals for areas that were not used for application					
Land application area totals	484	484	65		

B. CROPS AND HARVESTS

1

Field name: 1

1

11/14/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 80 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	162.00 *ton*	Dry-weight		45.4	37,600.00	3,900.00	29,500.00		10.70
05/19/2023	187.00 *ton*	Dry-weight		45.4	37,600.00	3,900.00	29,500.00		10.70
06/17/2023	210.00 *ton*	Dry-weight		45.4	37,600.00	3,900.00	29,500.00		10.70
07/14/2023	195.00 *ton*	Dry-weight		45.4	37,600.00	3,900.00	29,500.00		10.70
08/11/2023	225.00 *ton*	Dry-weight		43.2	35,800.00	3,800.00	37,800.00		12.80
09/07/2023	202.00 *ton*	Dry-weight		43.2	35,800.00	3,800.00	37,800.00		12.80
10/12/2023	173.00 *ton*	Dry-weight		43.2	35,800.00	3,800.00	37,800.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	16.93	692.00	72.52	625.67	2,191.81

10

Field name: 10

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	341.00 *ton*	Dry-weight		66.5	23,600.00	4,200.00	33,700.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.18	192.57	34.27	274.98	889.40

10

05/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 05/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	817.00 *ton*	Dry-weight		70.8	16,300.00	3,400.00	16,200.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.18	277.76	57.94	276.05	1,056.50

11

Field name: 11

11/16/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 17 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	27.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
05/19/2023	30.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
06/17/2023	36.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
07/14/2023	40.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
08/11/2023	47.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
09/07/2023	44.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
10/12/2023	32.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	15.06	639.87	68.06	589.74	2,142.15

12

Field name: 12

12

11/16/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 10 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	16.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
05/19/2023	19.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
06/17/2023	22.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
07/14/2023	23.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
08/11/2023	26.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
09/07/2023	24.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
10/12/2023	18.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.80	620.42	66.39	600.38	2,110.08

13

Field name: 13

11/16/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 14 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	23.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
05/19/2023	26.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
06/17/2023	30.00 *ton*	Dry-weight		41.4	38,700.00	4,000.00	27,400.00		12.00
07/14/2023	33.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
08/11/2023	37.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
09/07/2023	35.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30
10/12/2023	28.00 *ton*	Dry-weight		41.8	35,100.00	3,800.00	37,100.00		12.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	15.14	644.08	68.47	591.46	2,153.75

2

Field name: 2

11/15/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 12 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	22.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
05/19/2023	29.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
06/17/2023	32.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
07/14/2023	34.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
08/11/2023	38.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40
09/07/2023	31.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40
10/12/2023	26.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	17.67	806.17	83.35	682.76	2,399.56

3

Field name: 3

3

11/15/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 7 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	11.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
05/19/2023	14.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
06/17/2023	18.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
07/14/2023	17.00 *ton*	Dry-weight		42.0	40,100.00	4,300.00	30,400.00		11.20
08/11/2023	20.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40
09/07/2023	16.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40
10/12/2023	12.00 *ton*	Dry-weight		42.3	38,600.00	3,800.00	37,100.00		12.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	15.43	704.16	72.82	595.84	2,094.83

4

Field name: 4

11/15/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 40 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	64.00 *ton*	Dry-weight		43.3	40,300.00	4,400.00	31,200.00		12.00
05/19/2023	69.00 *ton*	Dry-weight		43.3	40,300.00	4,400.00	31,200.00		12.00
06/17/2023	75.00 *ton*	Dry-weight		43.3	40,300.00	4,400.00	31,200.00		12.00
07/14/2023	90.00 *ton*	Dry-weight		43.3	40,300.00	4,400.00	31,200.00		12.00
08/11/2023	98.00 *ton*	Dry-weight		40.7	36,600.00	3,800.00	36,500.00		12.90
09/07/2023	87.00 *ton*	Dry-weight		40.7	36,600.00	3,800.00	36,500.00		12.90
10/12/2023	71.00 *ton*	Dry-weight		40.7	36,600.00	3,800.00	36,500.00		12.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	13.85	618.28	66.02	540.64	1,992.96

5

Field name: 5

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 22 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	265.00 *ton*	Dry-weight		67.1	23,500.00	4,200.00	33,300.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.05	186.26	33.29	263.93	840.15

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 22 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	637.00 *ton*	Dry-weight		70.8	16,400.00	3,600.00	16,600.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.95	277.32	60.87	280.70	1,116.02

6

Field name: 6

11/14/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 68 Plant date: 11/14/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	815.00 *ton*	Dry-weight		67.4	22,700.00	4,000.00	33,900.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.99	177.39	31.26	264.91	851.77

6

05/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 68 Plant date: 05/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,905.00 *ton*	Dry-weight		69.9	15,500.00	3,400.00	16,200.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.01	261.41	57.34	273.21	1,113.08

7

Field name: 7

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 84 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	993.00 *ton*	Dry-weight		66.5	22,300.00	4,200.00	33,300.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.82	176.62	33.27	263.75	839.56

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 84 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	2,438.00 *ton*	Dry-weight		71.3	16,200.00	3,500.00	15,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.02	269.89	58.31	261.56	1,032.90

8

Field name: 8

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	272.00 *ton*	Dry-weight		67.5	21,900.00	4,400.00	34,400.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.83	168.34	33.82	264.43	837.88

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	632.00 *ton*	Dry-weight		71.1	15,800.00	3,400.00	15,200.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.48	250.94	54.00	241.41	952.95

9

Field name: 9

11/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 11/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	246.00 *ton*	Dry-weight		67.0	22,600.00	4,000.00	33,200.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.71	174.73	30.93	256.68	819.53

9

05/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 05/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	598.00 *ton*	Dry-weight		70.4	16,400.00	3,700.00	16,100.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.48	276.47	62.37	271.41	1,062.05

Fowler

Field name: Fowler

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	229.00 *ton*	Dry-weight		66.3	22,300.00	4,100.00	33,700.00		10.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.45	172.10	31.64	260.07	818.03

05/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 05/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	571.00 *ton*	Dry-weight		71.0	15,600.00	3,600.00	15,800.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.55	258.32	59.61	261.63	1,043.22

Harding

Field name: Harding

11/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 11/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/21/2023	432.00 *ton*	Dry-weight		66.5	22,000.00	4,000.00	32,600.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.37	167.57	30.47	248.31	792.15

05/09/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/09/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/01/2023	1,062.00 *ton*	Dry-weight		71.7	15,800.00	3,800.00	16,900.00		1.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.95	249.93	60.11	267.33	268.91

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/14/2022: Alfalfa, hay

Field name: 1

Crop: Alfalfa, hay

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	399.13	81.35	273.46	429.96	1,648,752.35 *gal*
Application event totals		399.13	81.35	273.46	429.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.43	12,325,050.00 *gal*
Application event totals		0.00	0.00	0.00	42.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.96	12,190,350.00 *gal*
Application event totals		0.00	0.00	0.00	41.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.62	12,089,325.00 *gal*
Application event totals		0.00	0.00	0.00	41.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.19	12,257,700.00 *gal*
Application event totals		0.00	0.00	0.00	42.19	

1 - 11/14/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.85	12,156,675.00 *gal*
Application event totals		0.00	0.00	0.00	41.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.50	12,055,650.00 *gal*
Application event totals		0.00	0.00	0.00	41.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.08	12,224,025.00 *gal*
Application event totals		0.00	0.00	0.00	42.08	

10 - 11/16/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.06	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.76	4,512,450.00 *gal*
Application event totals		0.00	0.00	0.00	49.76	

10 - 11/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	238.72	48.66	163.56	257.17	345,147.52 *gal*
Application event totals		238.72	48.66	163.56	257.17	

10 - 05/09/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 05/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.65	4,411,425.00 *gal*
Application event totals		0.00	0.00	0.00	48.65	

04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	303.94	196.35	240.81	815.05	621,528.36 *gal*
Application event totals		303.94	196.35	240.81	815.05	

06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.90	4,344,075.00 *gal*
Application event totals		0.00	0.00	0.00	47.90	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.02	4,445,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.02	

10 - 05/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.06	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	43.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.72	4,344,075.00 *gal*
Application event totals		0.00	0.00	0.00	42.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.02	4,445,100.00 *gal*
Application event totals		0.00	0.00	0.00	49.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.27	4,377,750.00 *gal*
Application event totals		0.00	0.00	0.00	48.27	

11 - 11/16/2022: Alfalfa, hay

Field name: 11

Crop: Alfalfa, hay

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	439.11	89.50	300.85	473.03	385,456.88 *gal*
Application event totals		439.11	89.50	300.85	473.03	

11 - 11/16/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.00	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	42.00	

04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.91	2,525,625.00 *gal*
Application event totals		0.00	0.00	0.00	40.91	

05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.55	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	42.55	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.37	2,491,950.00 *gal*
Application event totals		0.00	0.00	0.00	40.37	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.46	2,559,300.00 *gal*
Application event totals		0.00	0.00	0.00	41.46	

08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.10	2,592,975.00 *gal*
Application event totals		0.00	0.00	0.00	47.10	

11 - 11/16/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.55	2,626,650.00 *gal*
Application event totals		0.00	0.00	0.00	42.55	

12 - 11/16/2022: Alfalfa, hay

Field name: 12

Crop: Alfalfa, hay

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	456.30	93.00	312.63	491.56	235,618.22 *gal*
Application event totals		456.30	93.00	312.63	491.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.87	1,582,725.00 *gal*
Application event totals		0.00	0.00	0.00	48.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.91	1,616,400.00 *gal*
Application event totals		0.00	0.00	0.00	49.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.80	1,481,700.00 *gal*
Application event totals		0.00	0.00	0.00	40.80	

12 - 11/16/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	1,515,375.00 *gal*
Application event totals		0.00	0.00	0.00	41.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.83	1,549,050.00 *gal*
Application event totals		0.00	0.00	0.00	47.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.80	1,481,700.00 *gal*
Application event totals		0.00	0.00	0.00	40.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.91	1,616,400.00 *gal*
Application event totals		0.00	0.00	0.00	49.91	

13 - 11/16/2022: Alfalfa, hay

Field name: 13

Crop: Alfalfa, hay

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/07/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	417.19	85.03	285.83	449.42	301,587.95 *gal*
Application event totals		417.19	85.03	285.83	449.42	

13 - 11/16/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.27	2,188,875.00 *gal*
Application event totals		0.00	0.00	0.00	48.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.76	2,256,225.00 *gal*
Application event totals		0.00	0.00	0.00	49.76	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.02	2,222,550.00 *gal*
Application event totals		0.00	0.00	0.00	49.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.50	2,289,900.00 *gal*
Application event totals		0.00	0.00	0.00	50.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.27	2,188,875.00 *gal*
Application event totals		0.00	0.00	0.00	48.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.53	2,155,200.00 *gal*
Application event totals		0.00	0.00	0.00	47.53	

13 - 11/16/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.71	2,323,575.00 *gal*
Application event totals		0.00	0.00	0.00	45.71	

2 - 11/15/2022: Alfalfa, hay

Field name: 2

Crop: Alfalfa, hay

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	498.30	101.56	341.41	536.80	308,765.14 *gal*
Application event totals		498.30	101.56	341.41	536.80	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.05	1,919,475.00 *gal*
Application event totals		0.00	0.00	0.00	44.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.50	1,852,125.00 *gal*
Application event totals		0.00	0.00	0.00	42.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.96	1,784,775.00 *gal*
Application event totals		0.00	0.00	0.00	40.96	

2 - 11/15/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	40.96	1,784,775.00 *gal*
Application event totals		0.00	0.00	0.00	40.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	39.41	1,717,425.00 *gal*
Application event totals		0.00	0.00	0.00	39.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.50	1,852,125.00 *gal*
Application event totals		0.00	0.00	0.00	42.50	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	1,818,450.00 *gal*
Application event totals		0.00	0.00	0.00	41.73	

3 - 11/15/2022: Alfalfa, hay

Field name: 3

Crop: Alfalfa, hay

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	387.67	79.01	265.61	417.62	140,125.31 *gal*
Application event totals		387.67	79.01	265.61	417.62	

3 - 11/15/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.69	1,212,300.00 *gal*
Application event totals		0.00	0.00	0.00	47.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.04	1,144,950.00 *gal*
Application event totals		0.00	0.00	0.00	45.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.39	1,077,600.00 *gal*
Application event totals		0.00	0.00	0.00	42.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.69	1,212,300.00 *gal*
Application event totals		0.00	0.00	0.00	47.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.72	1,111,275.00 *gal*
Application event totals		0.00	0.00	0.00	43.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.39	1,077,600.00 *gal*
Application event totals		0.00	0.00	0.00	42.39	

3 - 11/15/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.69	1,212,300.00 *gal*
Application event totals		0.00	0.00	0.00	47.69	

4 - 11/15/2022: Alfalfa, hay

Field name: 4

Crop: Alfalfa, hay

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/06/2022	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	443.43	90.38	303.81	477.69	915,879.55 *gal*
Application event totals		443.43	90.38	303.81	477.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.12	6,263,550.00 *gal*
Application event totals		0.00	0.00	0.00	43.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.59	6,330,900.00 *gal*
Application event totals		0.00	0.00	0.00	43.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.89	6,229,875.00 *gal*
Application event totals		0.00	0.00	0.00	42.89	

4 - 11/15/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.35	6,297,225.00 *gal*
Application event totals		0.00	0.00	0.00	43.35	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.12	6,263,550.00 *gal*
Application event totals		0.00	0.00	0.00	43.12	

08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.89	6,229,875.00 *gal*
Application event totals		0.00	0.00	0.00	42.89	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.35	6,297,225.00 *gal*
Application event totals		0.00	0.00	0.00	43.35	

5 - 11/14/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.73	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	47.73	

5 - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.79	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	46.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	200.19	129.33	158.61	536.83	321,645.78 *gal*
Application event totals		200.19	129.33	158.61	536.83	

5 - 05/10/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.26	3,367,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	320.08	206.78	253.60	858.34	514,285.47 *gal*
Application event totals		320.08	206.78	253.60	858.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.68	3,468,525.00 *gal*
Application event totals		0.00	0.00	0.00	48.68	

5 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.73	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	47.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.32	3,300,150.00 *gal*
Application event totals		0.00	0.00	0.00	46.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.79	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	46.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.21	3,434,850.00 *gal*
Application event totals		0.00	0.00	0.00	48.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.15	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	49.15	

6 - 11/14/2022: Oats, silage-soft dough

Field name: 6

Crop: Oats, silage-soft dough

Plant date: 11/14/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

6 - 11/14/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/04/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.32	10,203,525.00 *gal*
Application event totals		0.00	0.00	0.00	41.32	

03/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.00	10,371,900.00 *gal*
Application event totals		0.00	0.00	0.00	42.00	

09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	213.01	137.61	168.76	571.20	1,057,843.52 *gal*
Application event totals		213.01	137.61	168.76	571.20	

6 - 05/10/2023: Corn, silage

Field name: 6

Crop: Corn, silage

Plant date: 05/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.18	10,169,850.00 *gal*
Application event totals		0.00	0.00	0.00	46.18	

04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	283.07	182.87	224.27	759.08	1,405,784.67 *gal*
Application event totals		283.07	182.87	224.27	759.08	

6 - 05/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.64	10,270,875.00 *gal*
Application event totals		0.00	0.00	0.00	46.64	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.32	10,203,525.00 *gal*
Application event totals		0.00	0.00	0.00	41.32	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	10,304,550.00 *gal*
Application event totals		0.00	0.00	0.00	41.73	

07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.48	10,237,200.00 *gal*
Application event totals		0.00	0.00	0.00	46.48	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.18	10,169,850.00 *gal*
Application event totals		0.00	0.00	0.00	46.18	

08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.73	10,304,550.00 *gal*
Application event totals		0.00	0.00	0.00	41.73	

7 - 11/15/2022: Oats, silage-soft dough

7 - 11/15/2022: Oats, silage-soft dough

Field name: 7

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.39	12,931,200.00 *gal*
Application event totals		0.00	0.00	0.00	42.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.17	12,863,850.00 *gal*
Application event totals		0.00	0.00	0.00	42.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	194.05	125.36	153.75	520.37	1,190,453.45 *gal*
Application event totals		194.05	125.36	153.75	520.37	

7 - 05/08/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.41	12,897,525.00 *gal*
Application event totals		0.00	0.00	0.00	47.41	

7 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	281.26	181.70	222.84	754.24	1,725,483.22 *gal*
Application event totals		281.26	181.70	222.84	754.24	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.39	12,931,200.00 *gal*
Application event totals		0.00	0.00	0.00	42.39	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.95	12,796,500.00 *gal*
Application event totals		0.00	0.00	0.00	41.95	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.62	12,695,475.00 *gal*
Application event totals		0.00	0.00	0.00	41.62	

07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.16	12,830,175.00 *gal*
Application event totals		0.00	0.00	0.00	47.16	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.28	12,863,850.00 *gal*
Application event totals		0.00	0.00	0.00	47.28	

7 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.04	12,796,500.00 *gal*
Application event totals		0.00	0.00	0.00	47.04	

8 - 11/16/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.53	3,468,525.00 *gal*
Application event totals		0.00	0.00	0.00	41.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.93	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	41.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	191.26	123.56	151.53	512.88	321,264.29 *gal*
Application event totals		191.26	123.56	151.53	512.88	

8 - 05/08/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 05/08/2023

8 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.92	3,569,550.00 *gal*
Application event totals		0.00	0.00	0.00	47.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	315.68	203.94	250.11	846.52	530,258.77 *gal*
Application event totals		315.68	203.94	250.11	846.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.56	3,468,525.00 *gal*
Application event totals		0.00	0.00	0.00	46.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.93	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	41.93	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.11	3,434,850.00 *gal*
Application event totals		0.00	0.00	0.00	46.11	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.47	3,535,875.00 *gal*
Application event totals		0.00	0.00	0.00	47.47	

8 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	41.53	3,468,525.00 *gal*
Application event totals		0.00	0.00	0.00	41.53	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.02	3,502,200.00 *gal*
Application event totals		0.00	0.00	0.00	47.02	

9 - 11/16/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 11/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.52	3,300,150.00 *gal*
Application event totals		0.00	0.00	0.00	48.52	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	44.16	3,367,500.00 *gal*
Application event totals		0.00	0.00	0.00	44.16	

09/15/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	205.90	133.02	163.13	552.14	315,784.17 *gal*
Application event totals		205.90	133.02	163.13	552.14	

9 - 05/08/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.04	3,434,850.00 *gal*
Application event totals		0.00	0.00	0.00	45.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/23/2023	Injection	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Slurry	Process wastewater	357.45	230.92	283.21	958.55	548,219.75 *gal*
Application event totals		357.45	230.92	283.21	958.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.02	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	49.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.03	3,266,475.00 *gal*
Application event totals		0.00	0.00	0.00	48.03	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.02	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	49.02	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	50.01	3,401,175.00 *gal*
Application event totals		0.00	0.00	0.00	50.01	

9 - 05/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.72	3,333,825.00 *gal*
Application event totals		0.00	0.00	0.00	43.72	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.28	3,300,150.00 *gal*
Application event totals		0.00	0.00	0.00	43.28	

Fowler - 11/15/2022: Oats, silage-soft dough

Field name: Fowler

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.12	3,131,775.00 *gal*
Application event totals		0.00	0.00	0.00	43.12	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	49.39	3,199,125.00 *gal*
Application event totals		0.00	0.00	0.00	49.39	

09/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	216.10	66.65	430.17	0.00	216.00 *ton*
Application event totals		216.10	66.65	430.17	0.00	

Fowler - 05/09/2023: Corn, silage

Field name: Fowler

Crop: Corn, silage

Plant date: 05/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	288.13	88.86	573.57	0.00	288.00 *ton*
Application event totals		288.13	88.86	573.57	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.66	3,098,100.00 *gal*
Application event totals		0.00	0.00	0.00	42.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.27	2,997,075.00 *gal*
Application event totals		0.00	0.00	0.00	46.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.79	3,030,750.00 *gal*
Application event totals		0.00	0.00	0.00	46.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	45.75	2,963,400.00 *gal*
Application event totals		0.00	0.00	0.00	45.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.31	3,064,425.00 *gal*
Application event totals		0.00	0.00	0.00	47.31	

Fowler - 05/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	42.66	3,098,100.00 *gal*
Application event totals		0.00	0.00	0.00	42.66	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	46.79	3,030,750.00 *gal*
Application event totals		0.00	0.00	0.00	46.79	

Harding - 11/15/2022: Oats, silage-soft dough

Field name: Harding

Crop: Oats, silage-soft dough

Plant date: 11/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/05/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.70	5,994,150.00 *gal*
Application event totals		0.00	0.00	0.00	48.70	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.16	5,926,800.00 *gal*
Application event totals		0.00	0.00	0.00	48.16	

09/02/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	221.15	68.21	440.24	0.00	420.00 *ton*
Application event totals		221.15	68.21	440.24	0.00	

Harding - 05/09/2023: Corn, silage

Field name: Harding

Crop: Corn, silage

Plant date: 05/09/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/22/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	284.34	87.69	566.02	0.00	540.00 *ton*
Application event totals		284.34	87.69	566.02	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.98	6,027,825.00 *gal*
Application event totals		0.00	0.00	0.00	48.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.43	5,960,475.00 *gal*
Application event totals		0.00	0.00	0.00	48.43	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.88	5,893,125.00 *gal*
Application event totals		0.00	0.00	0.00	47.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.44	5,994,150.00 *gal*
Application event totals		0.00	0.00	0.00	43.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	48.16	5,926,800.00 *gal*
Application event totals		0.00	0.00	0.00	48.16	

Harding - 05/09/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	43.20	5,960,475.00 *gal*
Application event totals		0.00	0.00	0.00	43.20	

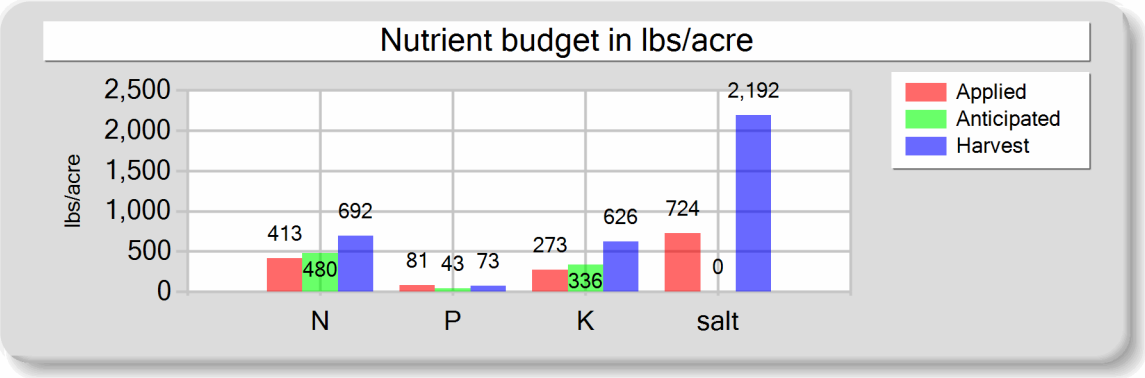
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Water - Canal	Surface water	0.00	0.00	0.00	47.88	5,893,125.00 *gal*
Application event totals		0.00	0.00	0.00	47.88	

B. NUTRIENT BUDGET

1 - 11/14/2022: Alfalfa, hay

Field name: 1 Crop: Alfalfa, hay Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	399.13	81.35	273.46	429.96
Fresh water	0.00	0.00	0.00	293.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	413.13	81.35	273.46	723.59
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	692.00	72.52	625.67	2,191.81
Nutrient balance	-278.87	8.83	-352.22	-1,468.23
Applied to removed ratio	0.60	1.12	0.44	0.33

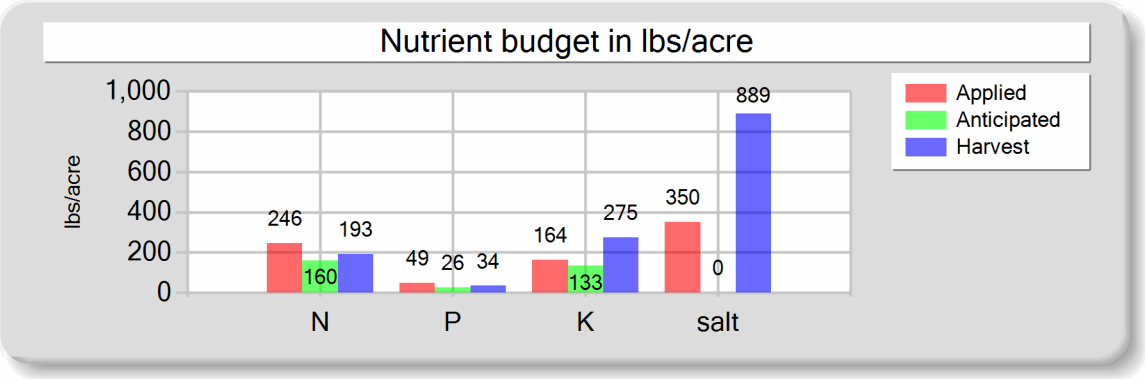
Fresh water applied	
85,298,775.00	*gallons*
3,141.26	*acre-inches*
39.27	*inches/acre*

Process wastewater applied	
1,648,752.35	*gallons*
60.72	*acre-inches*
0.76	*inches/acre*

Total harvests for the crop	
7	*harvests*

10 - 11/16/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.72	48.66	163.56	257.17
Fresh water	0.00	0.00	0.00	92.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	245.72	48.66	163.56	349.98
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	192.57	34.27	274.98	889.40
Nutrient balance	53.15	14.39	-111.42	-539.42
Applied to removed ratio	1.28	1.42	0.59	0.39

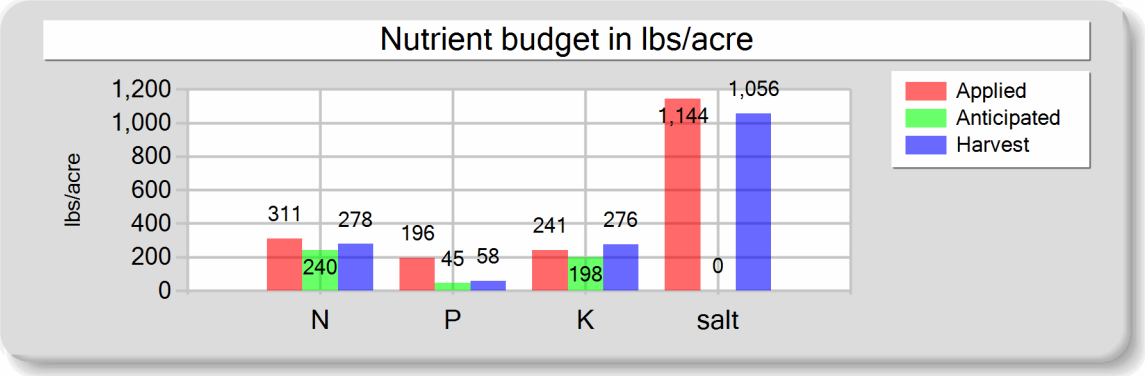
Fresh water applied	
8,890,200.00	*gallons*
327.40	*acre-inches*
11.69	*inches/acre*

Process wastewater applied	
345,147.52	*gallons*
12.71	*acre-inches*
0.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 05/09/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 05/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	303.94	196.35	240.81	815.05
Fresh water	0.00	0.00	0.00	328.64
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	310.94	196.35	240.81	1,143.69
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	277.76	57.94	276.05	1,056.50
Nutrient balance	33.18	138.41	-35.24	87.19
Applied to removed ratio	1.12	3.39	0.87	1.08

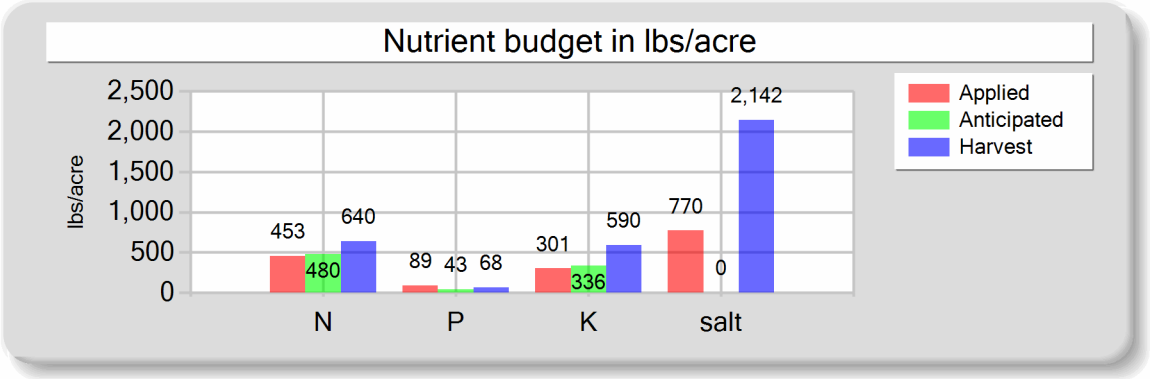
Fresh water applied	
30,745,275.00	*gallons*
1,132.24	*acre-inches*
40.44	*inches/acre*

Process wastewater applied	
621,528.36	*gallons*
22.89	*acre-inches*
0.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 11/16/2022: Alfalfa, hay

Field name: 11 Crop: Alfalfa, hay Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	439.11	89.50	300.85	473.03
Fresh water	0.00	0.00	0.00	296.94
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	453.11	89.50	300.85	769.97
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	639.87	68.06	589.74	2,142.15
Nutrient balance	-186.76	21.44	-288.89	-1,372.18
Applied to removed ratio	0.71	1.32	0.51	0.36

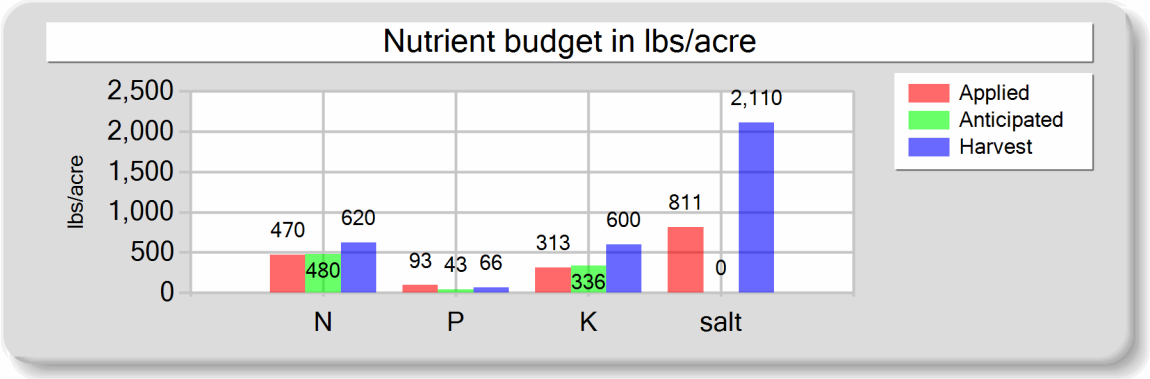
Fresh water applied	
18,016,125.00	*gallons*
663.47	*acre-inches*
39.03	*inches/acre*

Process wastewater applied	
385,456.88	*gallons*
14.20	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
7	*harvests*

12 - 11/16/2022: Alfalfa, hay

Field name: 12 Crop: Alfalfa, hay Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	456.30	93.00	312.63	491.56
Fresh water	0.00	0.00	0.00	319.85
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	470.30	93.00	312.63	811.41
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	620.42	66.39	600.38	2,110.08
Nutrient balance	-150.12	26.61	-287.75	-1,298.67
Applied to removed ratio	0.76	1.40	0.52	0.38

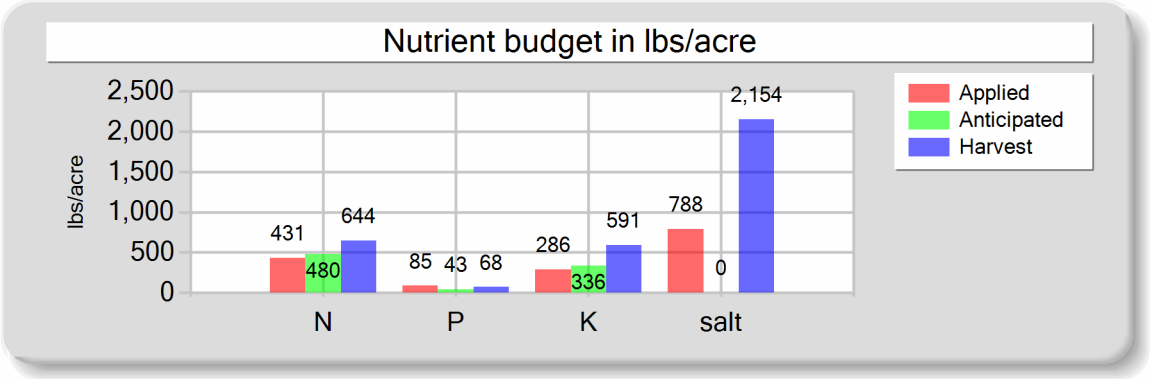
Fresh water applied	
10,843,350.00	*gallons*
399.32	*acre-inches*
39.93	*inches/acre*

Process wastewater applied	
235,618.22	*gallons*
8.68	*acre-inches*
0.87	*inches/acre*

Total harvests for the crop	
7	*harvests*

13 - 11/16/2022: Alfalfa, hay

Field name: 13 Crop: Alfalfa, hay Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	417.19	85.03	285.83	449.42
Fresh water	0.00	0.00	0.00	339.07
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	431.19	85.03	285.83	788.49
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	644.08	68.47	591.46	2,153.75
Nutrient balance	-212.89	16.56	-305.63	-1,365.26
Applied to removed ratio	0.67	1.24	0.48	0.37

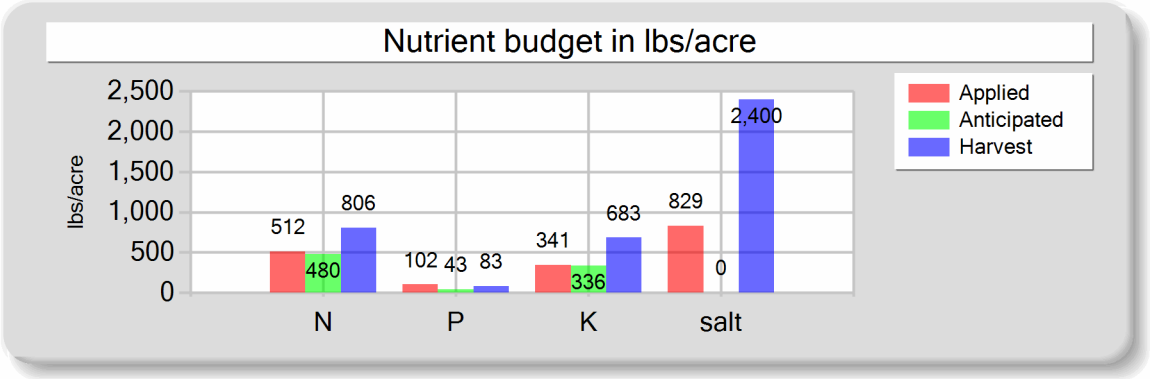
Fresh water applied	
15,625,200.00	*gallons*
575.42	*acre-inches*
41.10	*inches/acre*

Process wastewater applied	
301,587.95	*gallons*
11.11	*acre-inches*
0.79	*inches/acre*

Total harvests for the crop	
7	*harvests*

2 - 11/15/2022: Alfalfa, hay

Field name: 2 Crop: Alfalfa, hay Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	498.30	101.56	341.41	536.80
Fresh water	0.00	0.00	0.00	292.12
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	512.30	101.56	341.41	828.92
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	806.17	83.35	682.76	2,399.56
Nutrient balance	-293.87	18.21	-341.36	-1,570.64
Applied to removed ratio	0.64	1.22	0.50	0.35

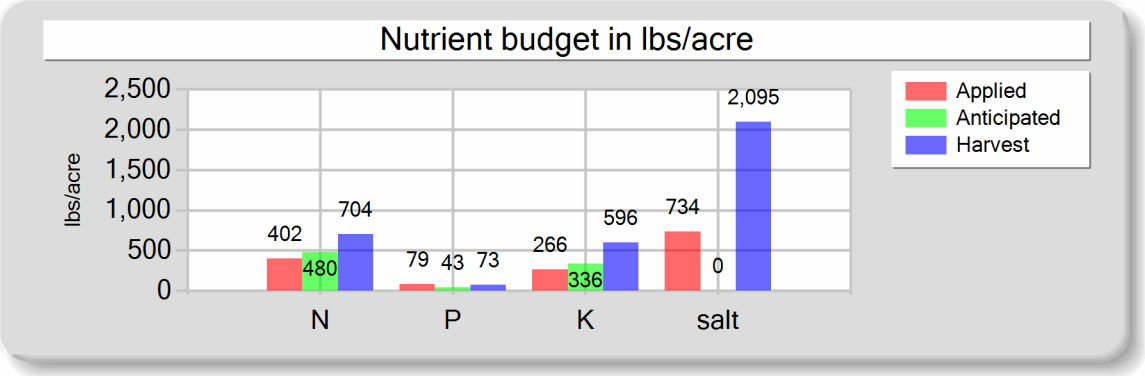
Fresh water applied	
12,729,150.00	*gallons*
468.77	*acre-inches*
39.06	*inches/acre*

Process wastewater applied	
308,765.14	*gallons*
11.37	*acre-inches*
0.95	*inches/acre*

Total harvests for the crop	
7	*harvests*

3 - 11/15/2022: Alfalfa, hay

Field name: 3 Crop: Alfalfa, hay Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	387.67	79.01	265.61	417.62
Fresh water	0.00	0.00	0.00	316.63
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	401.67	79.01	265.61	734.25
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	704.16	72.82	595.84	2,094.83
Nutrient balance	-302.48	6.19	-330.23	-1,360.58
Applied to removed ratio	0.57	1.09	0.45	0.35

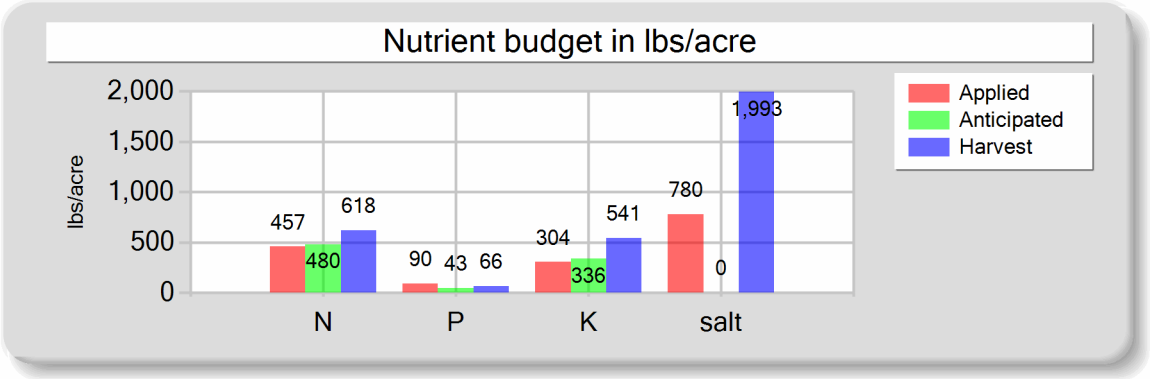
Fresh water applied	
8,048,325.00	*gallons*
296.39	*acre-inches*
42.34	*inches/acre*

Process wastewater applied	
140,125.31	*gallons*
5.16	*acre-inches*
0.74	*inches/acre*

Total harvests for the crop	
7	*harvests*

4 - 11/15/2022: Alfalfa, hay

Field name: 4 Crop: Alfalfa, hay Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	443.43	90.38	303.81	477.69
Fresh water	0.00	0.00	0.00	302.32
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	457.43	90.38	303.81	780.01
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	618.28	66.02	540.64	1,992.96
Nutrient balance	-160.85	24.36	-236.83	-1,212.95
Applied to removed ratio	0.74	1.37	0.56	0.39

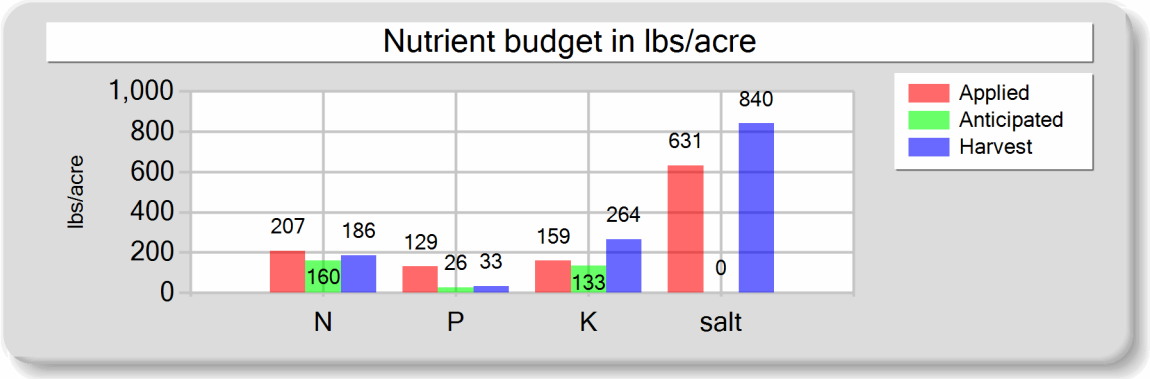
Fresh water applied	
43,912,200.00	*gallons*
1,617.14	*acre-inches*
40.43	*inches/acre*

Process wastewater applied	
915,879.55	*gallons*
33.73	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
7	*harvests*

5 - 11/14/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	200.19	129.33	158.61	536.83
Fresh water	0.00	0.00	0.00	94.52
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	207.19	129.33	158.61	631.35
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	186.26	33.29	263.93	840.15
Nutrient balance	20.93	96.04	-105.32	-208.80
Applied to removed ratio	1.11	3.88	0.60	0.75

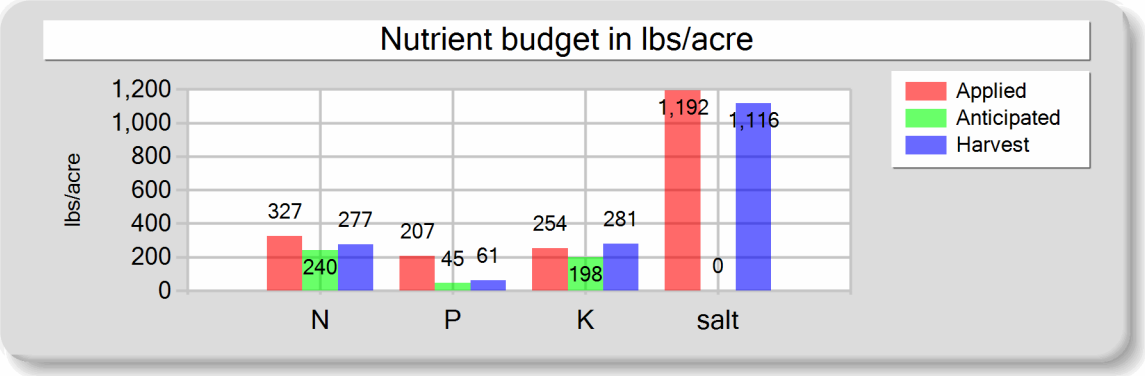
Fresh water applied	
6,735,000.00	*gallons*
248.03	*acre-inches*
11.27	*inches/acre*

Process wastewater applied	
321,645.78	*gallons*
11.85	*acre-inches*
0.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 05/10/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	320.08	206.78	253.60	858.34
Fresh water	0.00	0.00	0.00	334.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	327.08	206.78	253.60	1,192.49
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	277.32	60.87	280.70	1,116.02
Nutrient balance	49.77	145.91	-27.10	76.46
Applied to removed ratio	1.18	3.40	0.90	1.07

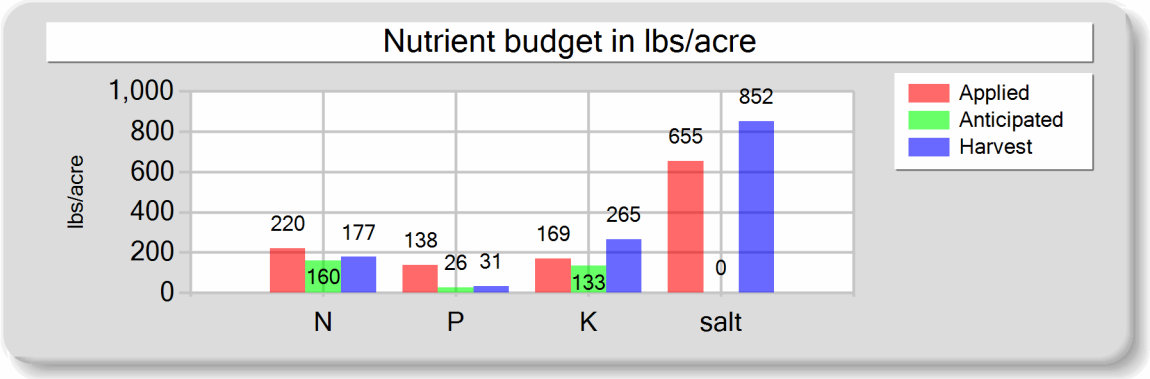
Fresh water applied	
23,808,225.00	*gallons*
876.78	*acre-inches*
39.85	*inches/acre*

Process wastewater applied	
514,285.47	*gallons*
18.94	*acre-inches*
0.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 11/14/2022: Oats, silage-soft dough

Field name: 6 Crop: Oats, silage-soft dough Plant date: 11/14/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	213.01	137.61	168.76	571.20
Fresh water	0.00	0.00	0.00	83.33
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	220.01	137.61	168.76	654.53
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	177.39	31.26	264.91	851.77
Nutrient balance	42.62	106.35	-96.14	-197.24
Applied to removed ratio	1.24	4.40	0.64	0.77

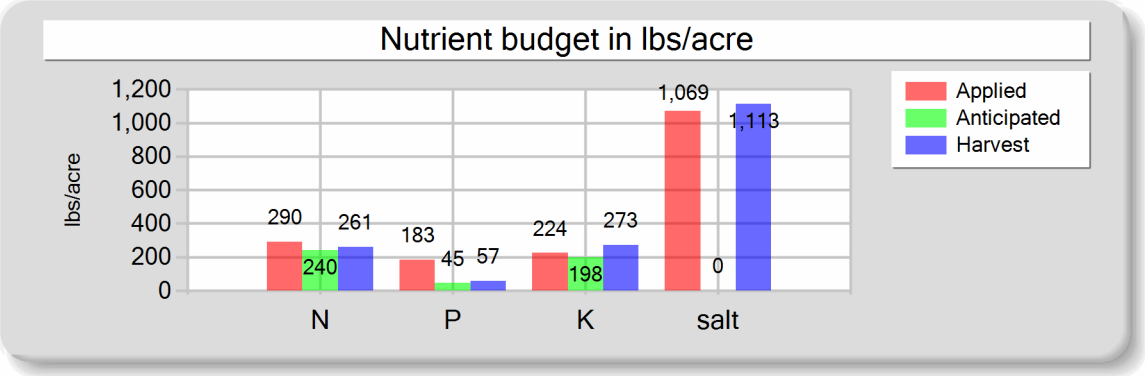
Fresh water applied	
20,575,425.00	*gallons*
757.72	*acre-inches*
11.14	*inches/acre*

Process wastewater applied	
1,057,843.52	*gallons*
38.96	*acre-inches*
0.57	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 05/10/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	283.07	182.87	224.27	759.08
Fresh water	0.00	0.00	0.00	310.26
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	290.07	182.87	224.27	1,069.34
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	261.41	57.34	273.21	1,113.08
Nutrient balance	28.66	125.53	-48.94	-43.74
Applied to removed ratio	1.11	3.19	0.82	0.96

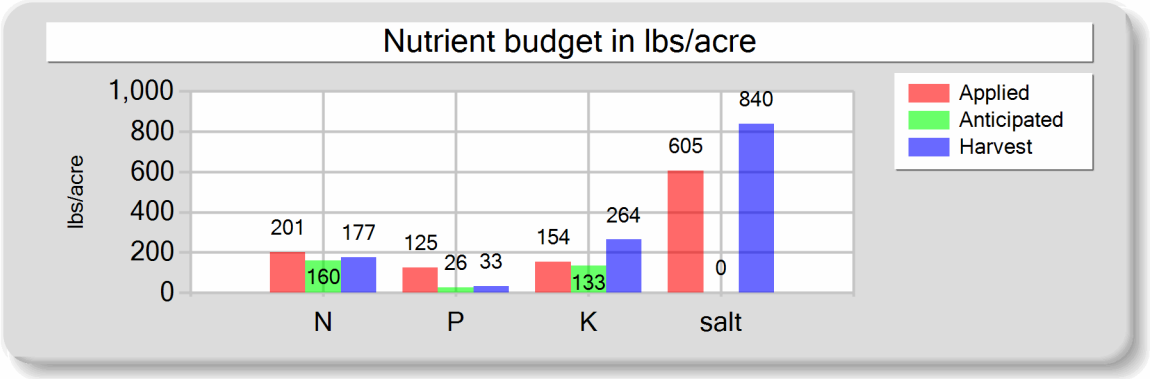
Fresh water applied	
71,660,400.00	*gallons*
2,639.01	*acre-inches*
38.81	*inches/acre*

Process wastewater applied	
1,405,784.67	*gallons*
51.77	*acre-inches*
0.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 11/15/2022: Oats, silage-soft dough

Field name: 7 Crop: Oats, silage-soft dough Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.05	125.36	153.75	520.37
Fresh water	0.00	0.00	0.00	84.57
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.05	125.36	153.75	604.94
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	176.62	33.27	263.75	839.56
Nutrient balance	24.43	92.10	-110.00	-234.62
Applied to removed ratio	1.14	3.77	0.58	0.72

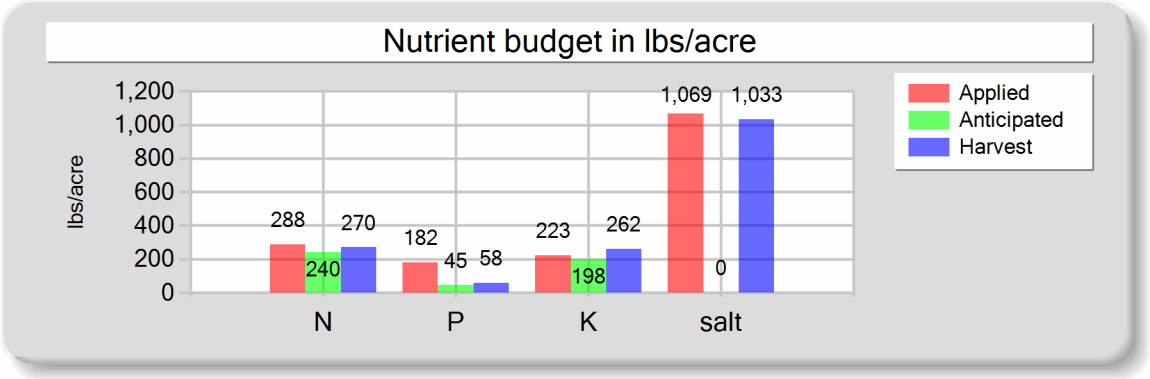
Fresh water applied	
25,795,050.00	*gallons*
949.94	*acre-inches*
11.31	*inches/acre*

Process wastewater applied	
1,190,453.45	*gallons*
43.84	*acre-inches*
0.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/08/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	281.26	181.70	222.84	754.24
Fresh water	0.00	0.00	0.00	314.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	288.26	181.70	222.84	1,069.10
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	269.89	58.31	261.56	1,032.90
Nutrient balance	18.38	123.39	-38.71	36.20
Applied to removed ratio	1.07	3.12	0.85	1.04

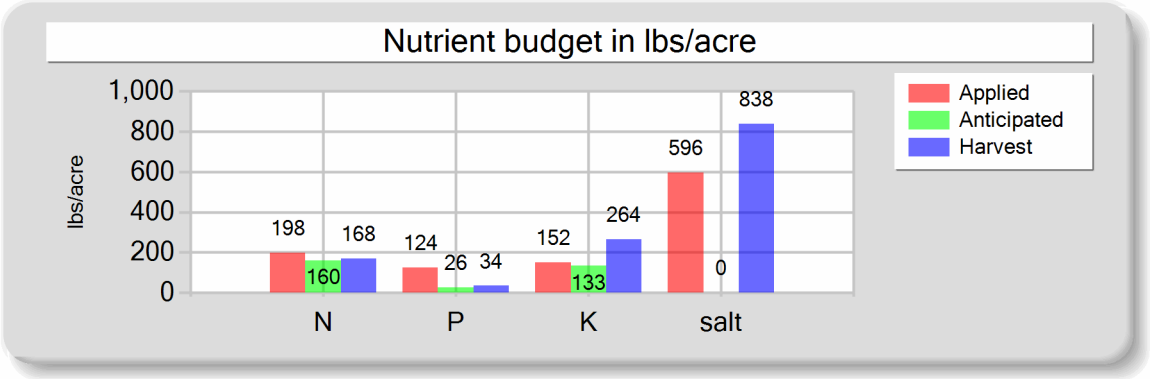
Fresh water applied	
89,811,225.00	*gallons*
3,307.44	*acre-inches*
39.37	*inches/acre*

Process wastewater applied	
1,725,483.22	*gallons*
63.54	*acre-inches*
0.76	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 11/16/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	191.26	123.56	151.53	512.88
Fresh water	0.00	0.00	0.00	83.46
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.26	123.56	151.53	596.34
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	168.34	33.82	264.43	837.88
Nutrient balance	29.91	89.73	-112.90	-241.54
Applied to removed ratio	1.18	3.65	0.57	0.71

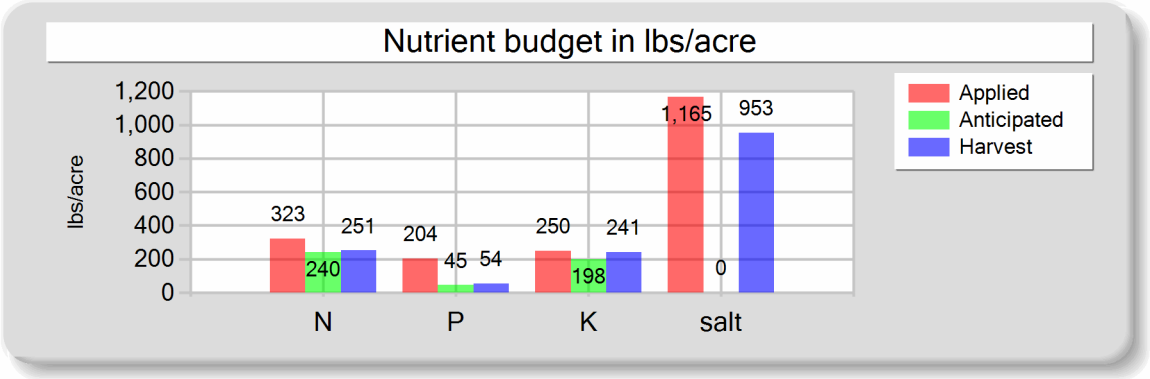
Fresh water applied	
6,970,725.00	*gallons*
256.71	*acre-inches*
11.16	*inches/acre*

Process wastewater applied	
321,264.29	*gallons*
11.83	*acre-inches*
0.51	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 05/08/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	315.68	203.94	250.11	846.52
Fresh water	0.00	0.00	0.00	318.54
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.68	203.94	250.11	1,165.06
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	250.94	54.00	241.41	952.95
Nutrient balance	71.73	149.93	8.70	212.12
Applied to removed ratio	1.29	3.78	1.04	1.22

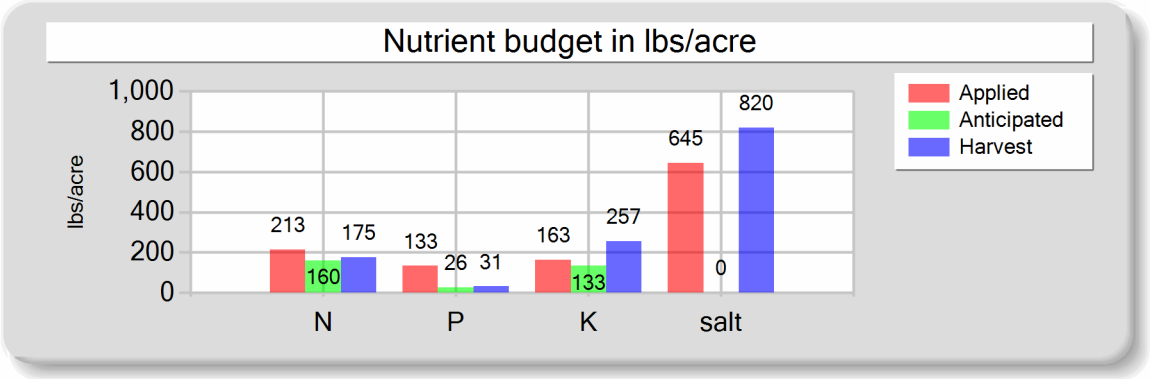
Fresh water applied	
24,481,725.00	*gallons*
901.58	*acre-inches*
39.20	*inches/acre*

Process wastewater applied	
530,258.77	*gallons*
19.53	*acre-inches*
0.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 11/16/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 11/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	205.90	133.02	163.13	552.14
Fresh water	0.00	0.00	0.00	92.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	212.90	133.02	163.13	644.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	174.73	30.93	256.68	819.53
Nutrient balance	38.17	102.09	-93.55	-174.71
Applied to removed ratio	1.22	4.30	0.64	0.79

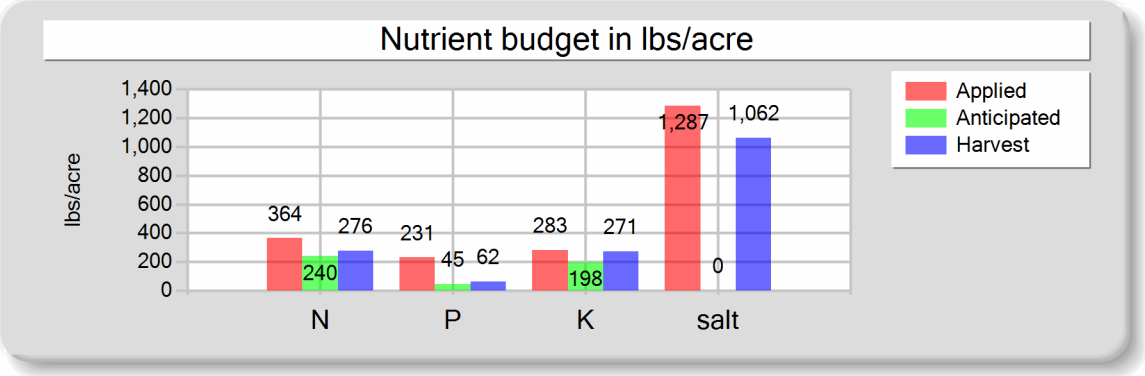
Fresh water applied	
6,667,650.00	*gallons*
245.55	*acre-inches*
11.69	*inches/acre*

Process wastewater applied	
315,784.17	*gallons*
11.63	*acre-inches*
0.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/08/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	357.45	230.92	283.21	958.55
Fresh water	0.00	0.00	0.00	328.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	364.45	230.92	283.21	1,286.66
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	276.47	62.37	271.41	1,062.05
Nutrient balance	87.98	168.55	11.80	224.61
Applied to removed ratio	1.32	3.70	1.04	1.21

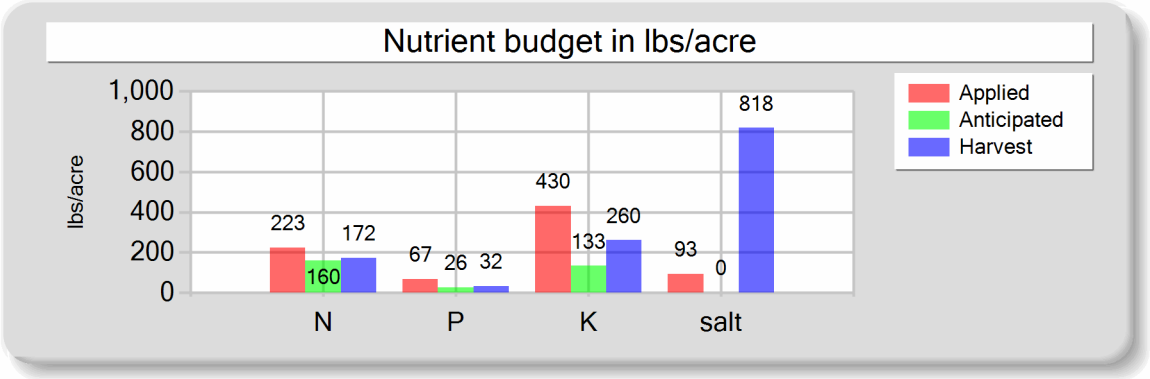
Fresh water applied	
23,404,125.00	*gallons*
861.89	*acre-inches*
41.04	*inches/acre*

Process wastewater applied	
548,219.75	*gallons*
20.19	*acre-inches*
0.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fowler - 11/15/2022: Oats, silage-soft dough

Field name: Fowler | Crop: Oats, silage-soft dough | Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	216.10	66.65	430.17	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	92.51
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	223.10	66.65	430.17	92.51
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.10	31.64	260.07	818.03
Nutrient balance	51.00	35.01	170.10	-725.52
Applied to removed ratio	1.30	2.11	1.65	0.11

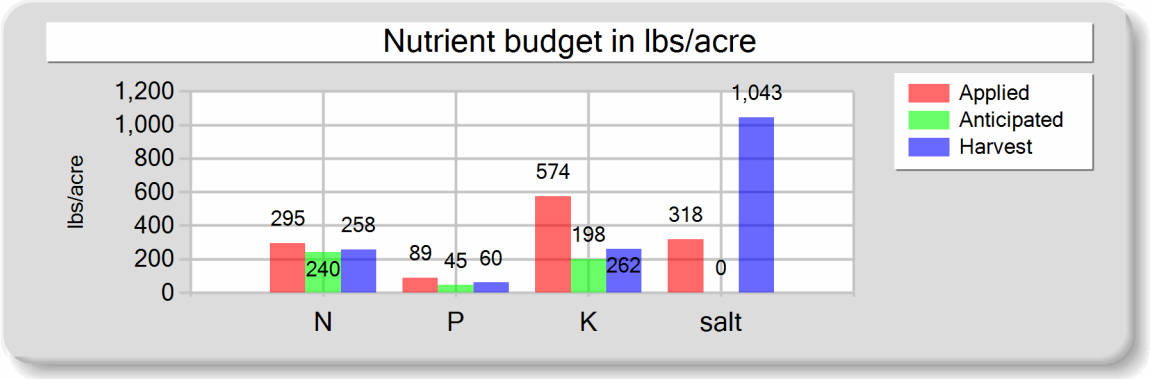
Fresh water applied	
6,330,900.00	*gallons*
233.15	*acre-inches*
11.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fowler - 05/09/2023: Corn, silage

Field name: Fowler | Crop: Corn, silage | Plant date: 05/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	288.13	88.86	573.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	318.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	295.13	88.86	573.57	318.22
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	258.32	59.61	261.63	1,043.22
Nutrient balance	36.81	29.25	311.93	-724.99
Applied to removed ratio	1.14	1.49	2.19	0.31

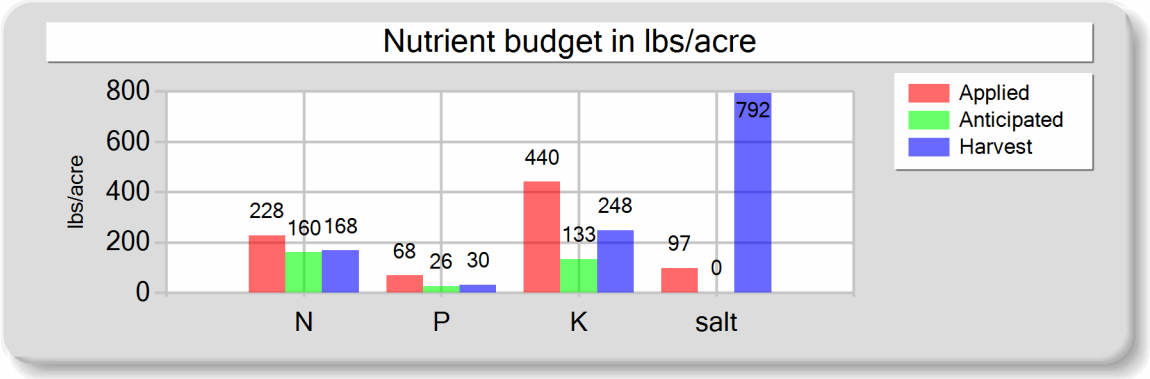
Fresh water applied	
21,282,600.00	*gallons*
783.77	*acre-inches*
39.19	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Harding - 11/15/2022: Oats, silage-soft dough

Field name: Harding | Crop: Oats, silage-soft dough | Plant date: 11/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	221.15	68.21	440.24	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	96.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	228.15	68.21	440.24	96.86
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	167.57	30.47	248.31	792.15
Nutrient balance	60.58	37.74	191.93	-695.29
Applied to removed ratio	1.36	2.24	1.77	0.12

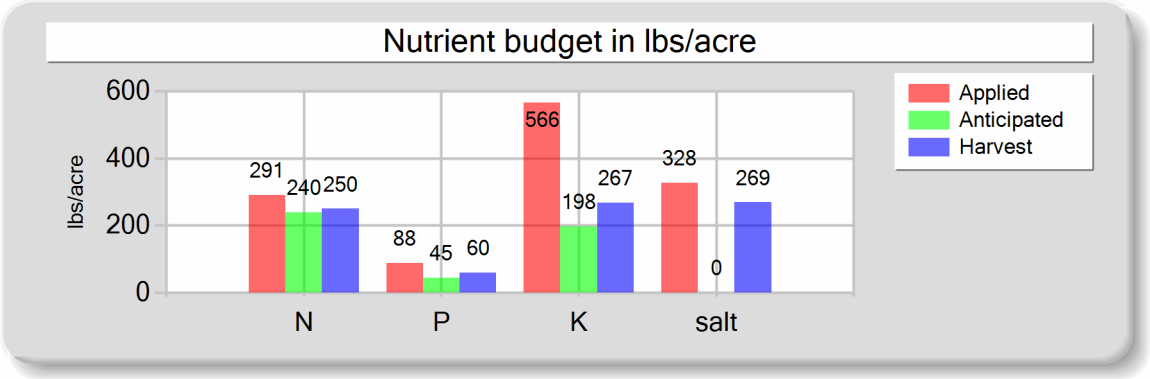
Fresh water applied	
11,920,950.00	*gallons*
439.01	*acre-inches*
11.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Harding - 05/09/2023: Corn, silage

Field name: Harding | Crop: Corn, silage | Plant date: 05/09/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.34	87.69	566.02	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	327.97
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	291.34	87.69	566.02	327.97
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	249.93	60.11	267.33	268.91
Nutrient balance	41.41	27.58	298.69	59.06
Applied to removed ratio	1.17	1.46	2.12	1.22

Fresh water applied	
41,655,975.00	*gallons*
1,534.05	*acre-inches*
40.37	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Solid Manure

Sample and source description: Solid Manure

Sample date: 05/24/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,700.00	3,300.00	21,300.00							
DL	100.00	100.00	100.00							

Fall Manure 2023

Sample and source description: Fall Manure 2023

Sample date: 11/01/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 6.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,500.00	3,100.00	16,200.00							0.00
DL	100.00	100.00	100.00							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water

Sample and source description: Lagoon Water

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,480.00	732.00	0.00	1.80	239.00	1,300.00								13,800.00	5,000
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Slurry

Sample and source description: Slurry

Sample date: 04/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,320.00	751.00	0.00	0.70	473.00	1,590.00								14,400.00	2,500
DL	1.00	0.50	0.10	0.10	0.10	0.50								10.00	10

Slurry

Sample and source description: Slurry

Sample date: 09/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,640.00	521.00	0.00	0.80	1,060.00	1,300.00								10,100.00	4,400
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

Q4 2023

Sample and source description: Q4 2023

Sample date: 11/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	102.00	38.30	0.00	0.70	33.40	331.00								1,870.00	1,550
DL	1.00	0.50	0.50	0.10	0.10	0.50								10.00	10

C. FRESH WATER ANALYSES

Domestic Well - DW #1

Domestic Well - DW #1

DW #1

Sample description: DW #1

Sample date: 11/20/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											1,700.00	
DL											10.00	

Irrigation Water - Canal

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										32.00	33
DL	0.01										10.00	10

Canal

Sample description: Canal

Sample date: 07/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00										33.50	37
DL	0.01										10.00	10

D. SOIL ANALYSES

1

1

1

Sample and source description: 1

Sample date: 09/08/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			2.00				

1

Sample and source description: 1

Sample date: 09/08/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			2.00				

10

10

Sample and source description: 10

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,114.00				
DL			2.00				

11

11

11

Sample and source description: 11

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,114.00				
DL			2.00				

12

12

Sample and source description: 12

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,114.00				
DL			2.00				

13

13

Sample and source description: 13

Sample date: 05/01/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			3,114.00				
DL			2.00				

2

2

2

Sample and source description: 2

Sample date: 09/08/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			92.00				
DL			2.00				

4

4

Sample and source description: 4

Sample date: 09/08/2022 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			94.00				
DL			2.00				

5

5

Sample and source description: 5

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,053.00				
DL			2.00				

6

6

6 EAST

Sample and source description: 6 EAST

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,083.00				
DL			2.00				

6 WEST

Sample and source description: 6 WEST

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,077.00				
DL			2.00				

7

7 EAST

Sample and source description: 7 EAST

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,243.00				
DL			2.00				

7

7 WEST

Sample and source description: 7 WEST

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,107.00				
DL			2.00				

8

8

Sample and source description: 8

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,708.00				
DL			2.00				

9

9

Sample and source description: 9

Sample date: 05/01/2021 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			2,297.00				
DL			2.00				

Fowler

Fowler

FWLR

Sample and source description: FWLR

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			210.00				
DL			2.00				

Harding

HARD E

Sample and source description: HARD E

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			213.00				
DL			2.00				

HARD W

Sample and source description: HARD W

Sample date: 09/06/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			188.00				
DL			2.00				

E. PLANT TISSUE ANALYSES

1 - 11/14/2022: Alfalfa, hay

1 - 11/14/2022: Alfalfa, hay

1A

Sample and source description: 1A

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,600.00	3,900.00	29,500.00		10.70
DL	100.00	100.00	100.00		0.01

1B

Sample and source description: 1B

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,700.00	4,300.00	29,400.00		11.30
DL	100.00	100.00	100.00		0.01

1A

Sample and source description: 1A

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,900.00	3,800.00	36,900.00		12.70
DL	100.00	100.00	100.00		0.01

1 - 11/14/2022: Alfalfa, hay

1B

Sample and source description: 1B

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,800.00	3,800.00	37,800.00		12.80
DL	100.00	100.00	100.00		0.01

10 - 11/16/2022: Oats, silage-soft dough

10

Sample and source description: 10

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	4,200.00	33,700.00		10.90
DL	100.00	100.00	100.00		0.01

10 - 05/09/2023: Corn, silage

10

Sample and source description: 10

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,300.00	3,400.00	16,200.00		6.20
DL	100.00	100.00	100.00		0.01

11 - 11/16/2022: Alfalfa, hay

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,700.00	4,000.00	27,400.00		12.00
DL	100.00	100.00	100.00		0.01

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,100.00	3,800.00	37,100.00		12.30
DL	100.00	100.00	100.00		0.01

12 - 11/16/2022: Alfalfa, hay

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,700.00	4,000.00	27,400.00		12.00
DL	100.00	100.00	100.00		0.01

12 - 11/16/2022: Alfalfa, hay

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,100.00	3,800.00	37,100.00		12.30
DL	100.00	100.00	100.00		0.01

13 - 11/16/2022: Alfalfa, hay

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,700.00	4,000.00	27,400.00		12.00
DL	1,000.00	100.00	1,000.00		0.01

11, 12, & 13

Sample and source description: 11, 12, & 13

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 41.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,100.00	3,800.00	37,100.00		12.30
DL	100.00	100.00	100.00		0.01

2 - 11/15/2022: Alfalfa, hay

2 - 11/15/2022: Alfalfa, hay

2 & 3

Sample and source description: 2 & 3

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	40,100.00	4,300.00	30,400.00		11.20
DL	100.00	100.00	100.00		0.01

2 & 3

Sample and source description: 2 & 3

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,600.00	3,800.00	37,100.00		12.40
DL	100.00	100.00	100.00		0.01

3 - 11/15/2022: Alfalfa, hay

2 & 3

Sample and source description: 2 & 3

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	40,100.00	4,300.00	30,400.00		11.20
DL	100.00	100.00	100.00		0.01

3 - 11/15/2022: Alfalfa, hay

2 & 3

Sample and source description: 2 & 3

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 42.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	38,600.00	3,800.00	37,100.00		12.40
DL	100.00	100.00	100.00		0.01

4 - 11/15/2022: Alfalfa, hay

4

Sample and source description: 4

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	40,300.00	4,400.00	31,200.00		12.00
DL	100.00	100.00	100.00		0.01

4

Sample and source description: 4

Sample date: 09/07/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 40.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	36,600.00	3,800.00	36,500.00		12.90
DL	100.00	100.00	100.00		0.01

5 - 11/14/2022: Oats, silage-soft dough

5 - 11/14/2022: Oats, silage-soft dough

5

Sample and source description: 5

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,500.00	4,200.00	33,300.00		10.60
DL	100.00	100.00	100.00		0.01

5 - 05/10/2023: Corn, silage

5

Sample and source description: 5

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,600.00	16,600.00		6.60
DL	100.00	100.00	100.00		0.01

6 - 11/14/2022: Oats, silage-soft dough

6A

Sample and source description: 6A

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	4,000.00	33,900.00		10.90
DL	100.00	100.00	100.00		0.01

6 - 11/14/2022: Oats, silage-soft dough

6B

Sample and source description: 6B

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	4,000.00	33,800.00		10.90
DL	100.00	100.00	100.00		0.01

6 - 05/10/2023: Corn, silage

6A

Sample and source description: 6A

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,600.00	3,600.00	16,200.00		6.60
DL	100.00	100.00	100.00		0.01

6B

Sample and source description: 6B

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,500.00	3,400.00	16,200.00		6.60
DL	100.00	100.00	100.00		0.01

7 - 11/15/2022: Oats, silage-soft dough

7 - 11/15/2022: Oats, silage-soft dough

7A

Sample and source description: 7A

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,600.00	4,000.00	34,200.00		10.60
DL	100.00	100.00	100.00		0.01

7B

Sample and source description: 7B

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	4,200.00	33,300.00		10.60
DL	100.00	100.00	100.00		0.01

7 - 05/08/2023: Corn, silage

7A

Sample and source description: 7A

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	3,600.00	16,200.00		5.70
DL	100.00	100.00	100.00		0.01

7 - 05/08/2023: Corn, silage

7B

Sample and source description: 7B

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	3,500.00	15,700.00		6.20
DL	100.00	100.00	100.00		0.01

8 - 11/16/2022: Oats, silage-soft dough

8

Sample and source description: 8

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,900.00	4,400.00	34,400.00		10.90
DL	100.00	100.00	100.00		0.01

8 - 05/08/2023: Corn, silage

8

Sample and source description: 8

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	3,400.00	15,200.00		6.00
DL	100.00	100.00	100.00		0.01

9 - 11/16/2022: Oats, silage-soft dough

9

Sample and source description: 9

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	4,000.00	33,200.00		10.60
DL	100.00	100.00	100.00		0.01

9 - 05/08/2023: Corn, silage

9

Sample and source description: 9

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,400.00	3,700.00	16,100.00		6.30
DL	100.00	100.00	100.00		0.01

Fowler - 11/15/2022: Oats, silage-soft dough

Fowler

Sample and source description: Fowler

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,300.00	4,100.00	33,700.00		10.60
DL	100.00	100.00	100.00		0.01

Fowler - 05/09/2023: Corn, silage

Fowler

Sample and source description: Fowler

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,600.00	3,600.00	15,800.00		6.30
DL	100.00	100.00	100.00		0.01

Harding - 11/15/2022: Oats, silage-soft dough

Harding

Sample and source description: Harding

Sample date: 04/21/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	4,000.00	32,600.00		10.40
DL	100.00	100.00	100.00		0.01

Harding - 05/09/2023: Corn, silage

Harding

Sample and source description: Harding

Sample date: 09/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	3,800.00	16,900.00		1.70
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

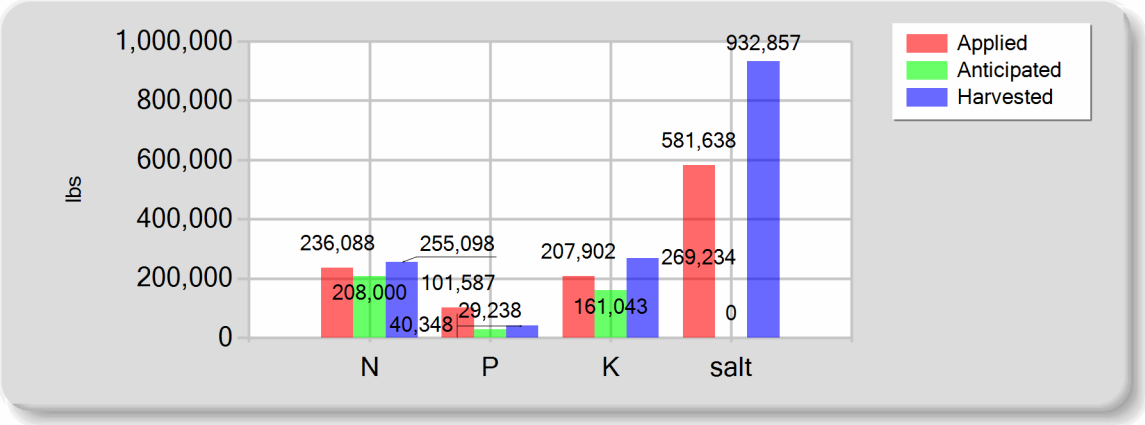
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

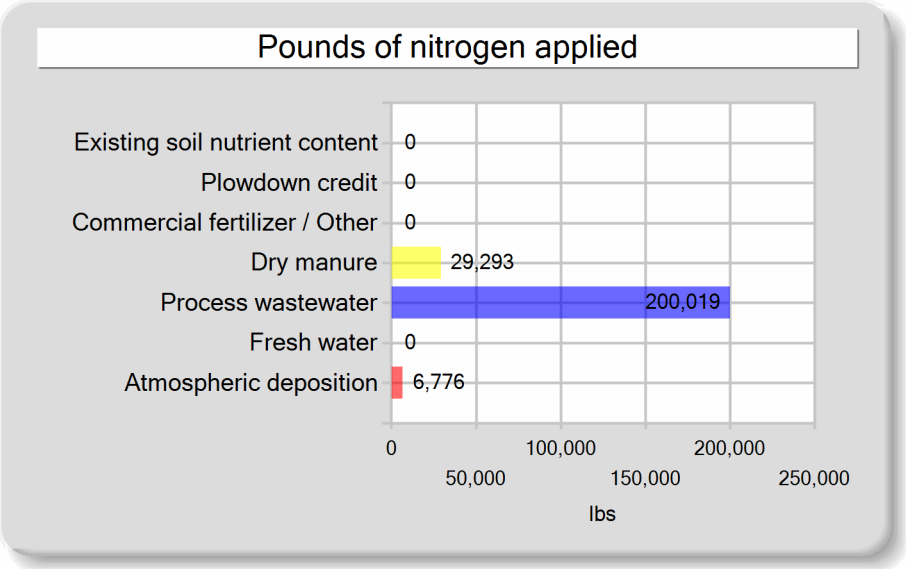
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

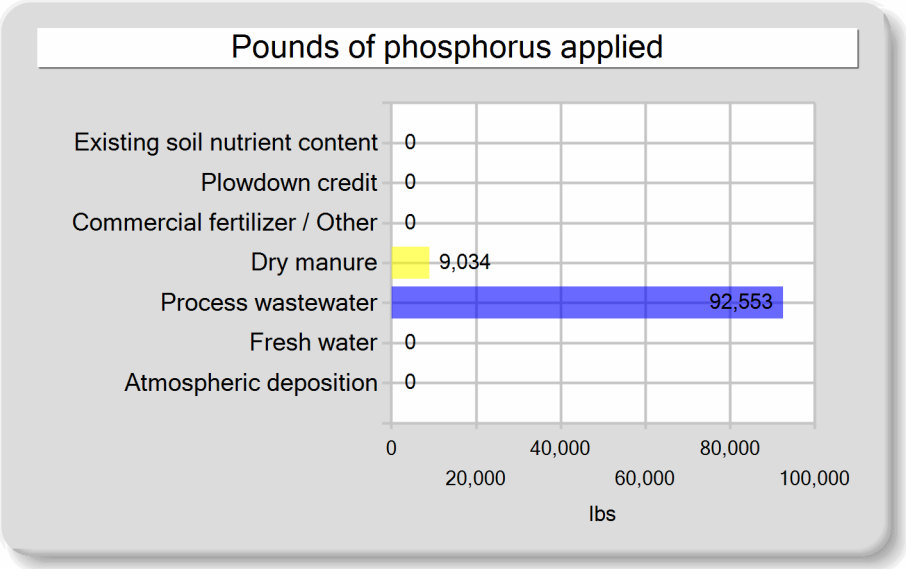
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	29,293.18	9,034.34	58,312.58	0.00
Process wastewater	200,018.93	92,552.52	149,589.43	403,351.89
Fresh water	0.00	0.00	0.00	178,285.89
Atmospheric deposition	6,776.00	0.00	0.00	0.00
Total nutrients applied	236,088.11	101,586.86	207,902.02	581,637.78
Anticipated crop nutrient removal	208,000.00	29,238.40	161,043.20	0.00
Actual crop nutrient removal	255,097.67	40,347.83	269,234.31	932,856.54
Nutrient balance	-19,009.56	61,239.03	-61,332.29	-351,218.76
Applied to removed ratio	0.93	2.52	0.77	0.62

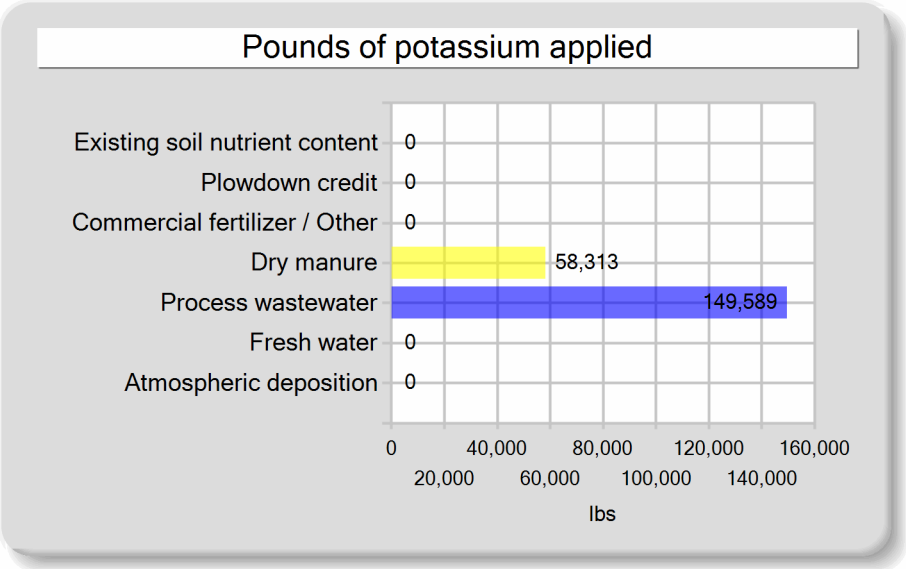
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

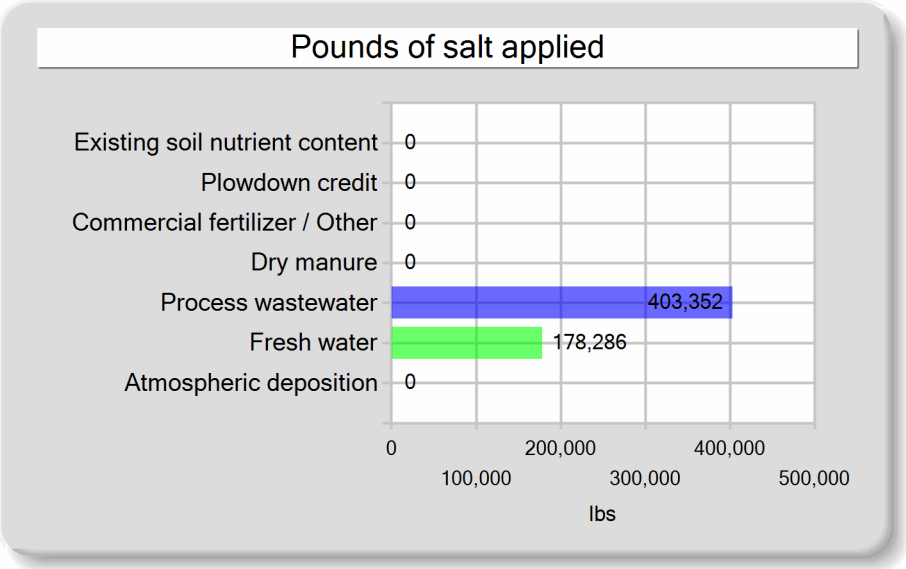


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

SOIL ANALYSES: Soluble phosphorus reported as...
*Phosphorus (Olsen Method):Fields 1, 2, 4
*Phosphates: Fields 5, 8, 7, 9, 6, 10, 11, 12, 13
*Phosphorus Weak Bray: Fields Fowler, Harding

*ALFALFA on Fields 1, 2, 3, 4, 11, 12 & 13 was harvested as SILAGE instead of Hay.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Bobby Borba
Bobby Borba (Jun 19, 2024 15:38 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Bob Borba	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
06/18/2024	
DATE	DATE

DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:28

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23K1077-01	DW #1	Ag Water	T. Madruga		11/20/2023 16:00

Default Cooler Temperature on Receipt °C: 0.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

R & J Avila Inc
8119 Ellington Ave
Hilmar, CA 95324

Account# 00-0019840
Account Manager: Ben Nydam
Submitted By: Rui & John Avila
Ranch: Dairy Central

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:28

Sample Results

Sample: DW #1
23K1077-01 (Water)

Sampled: 11/20/2023 16:00
Sampled By: T. Madruga

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.70**	mmhos/cm	0.01	1		11/28/23 15:09	SM 2510 B		BEK0749
Electrical Conductivity umhos	**1700**	umhos/cm	10.0	1		11/28/23 15:09	SM 2510 B		BEK0749
Ammonia (as N)	ND	mg/L	0.00	1		11/20/23 16:00	Field		BEK0702
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	11/22/23 02:40	EPA 300.0		BEK0692
Temperature	**25.0**	units	0.0	1		11/28/23 15:09	SM 4500-H+	H	BEK0749
pH	**7.9**	units	1.0	1		11/28/23 15:09	SM 4500-H+	H	BEK0749

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Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0692										
Blank (BEK0692-BLK1)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0692-BLK2)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEK0692-BLK3)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEK0692-BS1)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000		105	90-110		
LCS (BEK0692-BS2)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000		106	90-110		
Duplicate (BEK0692-DUP1)		**Source: 23K1075-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	0.1		0.1	mg/L		0.1			2.76	10
Duplicate (BEK0692-DUP2)		**Source: 23K1076-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L		5.1			0.452	10
Matrix Spike (BEK0692-MS1)		**Source: 23K1075-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L	5.000	0.1	101	90-110		
Matrix Spike (BEK0692-MS2)		**Source: 23K1076-02**			Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3		0.1	mg/L	5.000	5.1	103	90-110		
Reference (BEK0692-SRM1)					Prepared & Analyzed: 11/21/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0692-SRM2)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEK0692-SRM3)					Prepared: 11/21/2023 Analyzed: 11/22/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Ranch: Dairy Central

Received: 11/21/2023 16:09
Reported: 11/29/2023 11:28

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0749										
Blank (BEK0749-BLK1)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.5		1.0	units						
Blank (BEK0749-BLK2)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.0		1.0	units						
Blank (BEK0749-BLK3)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.5		1.0	units						
Duplicate (BEK0749-DUP1)		**Source: 23K1078-01**			Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1.17		0.01	mmhos/cm		1.20			2.76	10
pH	8.0		1.0	units		8.1			0.248	10
Electrical Conductivity umhos	1170		10.0	umhos/cm		1200			2.76	10
Duplicate (BEK0749-DUP2)		**Source: 23K1126-02**			Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1.82		0.01	mmhos/cm		1.81			0.727	10
Electrical Conductivity umhos	1820		10.0	umhos/cm		1810			0.727	10
pH	8.0		1.0	units		8.0			0.125	10
Reference (BEK0749-SRM1)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	552			umhos/cm	538.0		103	90-110		
Reference (BEK0749-SRM2)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEK0749-SRM3)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEK0749-SRM4)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEK0749-SRM5)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEK0749 (Continued)										
Reference (BEK0749-SRM5)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEK0749-SRM6)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0749-SRM7)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEK0749-SRM8)					Prepared: 11/22/2023	Analyzed: 11/28/2023				
pH	4.0			units	4.000		101	97.5-102.5		

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11/21/23 16:09 23K1077

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 1 No. of Bottles:

Purchase Order No | Bill To: | Acct # 19840 | Cons # 08

Water Type: [ ] Drinking Water [ ] Wastewater
[X] Ag Water [ ] Groundwater [ ] Monitoring Well

Results Need By

Name: R & J Avila Inc
Address: 8119 Ellington Avenue
City: Hilmar **State:** CA **Zip:** 95324
Telephone: (209) 669-7919 **Fax:** (209) 634-4604
Cell/Email: 209 678-3086 Rui / 209 678-1723 John frontdesk.randjavila@gmail.com

COPY TO:

REQUESTED BY: Rui & John Avila
PROJECT Dairy Central
ID CROP: N/A

[X] Copy of Chain [X] QA/QC Documents
Sampled By: T. Madruga

Other:

Analysis and Bottles Required: (Please indicate Analysis)

(X) DWW1: EC, pH, NO_3-N (NH4-N Field Test)
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DWW2: DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DCW1: EC, NO_3-N, TDS + TKN & TN
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW1: EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) DPW2: DPW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na
(1-1 Liter Plastic, Unpreserved) White Per Sample
( ) Other

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field NH_4-N
1	DW #1	11/20/2023	1600	0.9	ND
2					
3					
4					
5					
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	John Avila	R & J Avila, Inc.	11/20/2023 @ 1600	11/21/2023 @ 900
Second	JLF	Dellavalle Laboratory, Inc.	11/21/2023 @ 900	11-21-23 3:48
Third				
Fourth		DLI	11/21/23 16:09	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through ***cal*** under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration,reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Samples recieved in cooler with ice (coolant)
[ ] Yes [ ] No

11/21/23 16:09 23K1077

Shipping Information: Fed Ex ☐ UPS ☐ GSO ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐
Ontrac ☐ PMS ☐ USPS ☐ Other ☐ _____

Shipping Container: Ice Chest ☒ None ☐ Box ☐

Refrigerant: Wet Ice: ☒ Blue Ice: ☐ None: ☐

Container(s) Received	1	2	3	4	5	6	7	8	9	10
Plastics:										
100 mL sterile plastic $Na_2S_2O_3$ (Green)										
250 mL unpreserved (White)										
250 mL HNO_3 (Red)										
250 mL H_2SO_4 (Yellow)										
500 mL unpreserved (White)										
500 mL HNO_3 (Red)										
500 mL H_2SO_4 (Yellow)										
1 L unpreserved (White)	1									
1 L unpreserved (BOD) (Purple)										
1 L HNO_3 (Red)										
1 L H_2SO_4 (Yellow)										
VOA Vials in sets of 3:										
40 mL VOA, HCl (Blue)										
40 mL VOA, $Na_2S_2O_3$ (Green)										
40 mL VOA, unpreserved (White)										
Glass:										
250 mL AG unpreserved (White)										
250 mL AG H_2SO_4 (Yellow)										
250 mL AG $Na_2S_2O_3$ (Green)										
250 mL AG $Na_2S_2O_3$ + MCAA										
500 mL glass unpreserved (White)										
500 mL AG HCl (Blue)										
1 L AG unpreserved (White)										
1 L AG H_2SO_4 (Yellow)										
1 L AG $Na_2S_2O_3$ (Green)										
1 L AG HCl (Blue)										
Special:										
Cr^{6+} w/Borate/HCO3/CO3										
Cyanide - 500 mL NaOH										
Asbestos - 2 - 1L P wrapped in foil										
500 mL or 1 L **AP** $Na_2S_2O_3$ (Green)										
1 L **AG** or **P** NaOH + ZnAc (Sulfide)										
Chlorite/Bromate 250 mL AG with EDA										
TTHM										
HAA5										
40 mL VOA, $Na_2S_2O_3$ + MCAA										